

JavaScript Basics Assessment

Question 1 — let · var · const

Part A

Which statement about const is correct?

- A. A const variable can be reassigned at any time.
- B. const variables are function-scoped.
- C. A const variable must be assigned a value when declared.
- D. const prevents any change to an object's properties.

Part B

Fill in the blanks with let, var, or const:

- Use _____ when the value will never change.
- Use _____ when the value needs to change inside a block.

Part C

What will this code print?

```
var x = 10;
if (true) {
  var x = 20;
  let y = 30;
}
console.log(x);
console.log(y);
```

Question 2 — if · else if · nested if-else

Part A

What is printed when score = 55?

```
if (score >= 90) { console.log("A"); }
else if (score >= 75) { console.log("B"); }
else if (score >= 50) { console.log("C"); }
else { console.log("F"); }
```

- A. A
- B. B
- C. C
- D. F

Part B

Write a nested if-else for these age categories:

```
age < 13 → print "child"
age < 18 → print "teenager"
age < 60 → print "adult"
otherwise → print "senior"
```

Question 3 — Ternary operator

Part A

What is the value of result?

```
let num = 8;
let result = (num % 2 === 0) ? "even" : "odd";
console.log(result);
```

Part B

Rewrite this as a single ternary expression stored in a variable called access:

```
if (loggedIn) { access = "welcome!"; }
else { access = "please log in"; }
```

Question 4 — for · while · do-while

Part A

Which loop always runs at least once, even if the condition is false?

- A. for loop
- B. while loop
- C. do-while loop
- D. None of the above

Part B

What does this loop print?

```
let i = 1;
while (i <= 5) {
  if (i % 2 !== 0) { console.log(i); }
  i++;
}
```

Part C

Write a for loop that prints numbers from 10 down to 1.